

Case 2: Structural Vector AutoRegressive Models, Local Projections and Macro-economic Policy Analysis

EBC4008 – Time Series Analytics

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Recap – Part I

- Reduced-form VAR(14) was estimated for

$$\Delta \log(INDPRO_t), \quad \Delta^2 \log(CPI_t), \quad \Delta FFR_t$$

- Reduced-form IRFs were computed over 48 months with bootstrap confidence intervals
 - The responses were accumulated to recover the variables in levels
 - A positive FFR innovation is followed by an increase in the CPI level, producing a price-puzzle-like pattern
- $\Rightarrow$ *FFR innovation is **not yet structurally identified**, motivating the **SVAR** analysis*

Structural VARs

Reduced-Form VAR versus Structural VAR

- Reduced-form VAR: statistical dynamics and forecast errors

$$y_t = A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$$

- Structural VAR: contemporaneous relationships and economic shocks

$$B_0 y_t = B_1 y_{t-1} + \dots + B_p y_{t-p} + w_t$$

$$A_i = B_0^{-1} B_i, \quad u_t = B_0^{-1} w_t, \quad E(w_t w_t') = I_K$$

- u_t may combine several underlying economic disturbances
- B_0 captures contemporaneous relationships; w_t is the vector of structural shocks

Objective: Recover economically interpretable shocks from the reduced-form errors

From Reduced-Form to Structural VAR

- Premultiplying the reduced-form companion VAR by $\mathcal{B}_0$ yields the structural representation

$$\mathcal{B}_0 Y_t = \underbrace{\mathcal{B}_0 A}_{\mathcal{B}_1} Y_{t-1} + \underbrace{\mathcal{B}_0 U_t}_{W_t}$$

where $Y_t, U_t, W_t \in \mathbb{R}^{Kp}$, $A, \mathcal{B}_0, \mathcal{B}_1 \in \mathbb{R}^{Kp \times Kp}$, and $\mathcal{B}_0 = \text{diag}(B_0, I_K, \dots, I_K)$.

- Both representations describe the same dynamics, but the structural interpretation depends on identifying B_0

Identification Problem

- The reduced-form covariance matrix provides information about B_0^{-1}

$$\Sigma_u = B_0^{-1}(B_0^{-1})'$$

- However, it provides fewer independent equations than unknown coefficients

$$\underbrace{K^2}_{\text{Unknowns}} - \underbrace{\frac{K(K+1)}{2}}_{\text{Equations}} = \underbrace{\frac{K(K-1)}{2}}_{\text{Restrictions}}$$

- For the trivariate VAR:

$$9 - 6 = \boxed{3 \text{ restrictions}}$$

⇒ At least three independent restrictions are needed; the rank condition must also hold

Types of Identification Restrictions

- Structural restrictions can constrain the effects of shocks at different horizons
 - **Short-run:** zero, equality or proportionality restrictions on contemporaneous coefficients
 - **Long-run:** restrictions on cumulative or permanent effects of structural shocks
 - **Sign:** restrictions on the direction of responses over selected horizons
- Economic justification may arise from theory, information delays, physical constraints or institutional knowledge
- Here: Short-run exclusion restrictions to identify monetary-policy shocks

Impulse Response Functions for Recursive SVARs

Recursive Identification through Ordering

- Recursive identification imposes a lower-triangular structural impact matrix

$$u_t = \underbrace{\begin{bmatrix} b_{11}^0 & 0 & 0 \\ b_{21}^0 & b_{22}^0 & 0 \\ b_{31}^0 & b_{32}^0 & b_{33}^0 \end{bmatrix}}_{B_0^{-1}} w_t$$

- The three zeros satisfy the order condition for $K = 3$; the Cholesky decomposition selects the corresponding lower-triangular impact matrix

- Structural impulse responses are obtained from the reduced-form responses

$$\Theta_h = \Phi_h B_0^{-1}, \quad h = 0, \dots, H$$

- The ordering determines which shocks can affect each variable contemporaneously

Ordering the Monetary-Policy VAR

- We adopt the ordering

$$y_t = (IP_t, CPI_t, FFR_t)'$$

- Economic motivation:
 - **IP and CPI:** assumed not to respond within the month to unexpected monetary-policy changes
 - **FFR:** allowed to react contemporaneously to real activity and price developments
- The last innovation is interpreted as the monetary-policy shock

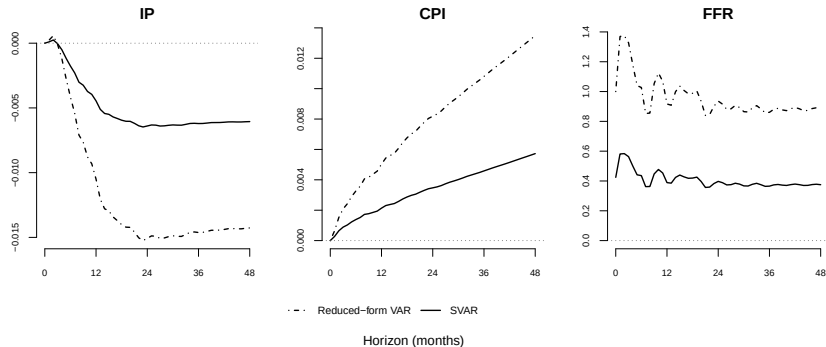
$$w_{3t} \equiv w_t^{MP}$$

⇒ Only the policy shock is assigned an economic interpretation (semistructural SVAR)

Estimating Recursive SVAR IRFs

- We use `SVAR()` with recursive zero restrictions on the structural impact matrix
 - `irf()` computes structural responses from the `VAR(14)` over 48 months
 - Structural shocks have unit variance and are oriented to raise FFR on impact
 - Responses are accumulated to recover levels
- ⇒ The figures compare point estimates across identification assumptions

Structural Monetary-Policy IRFs



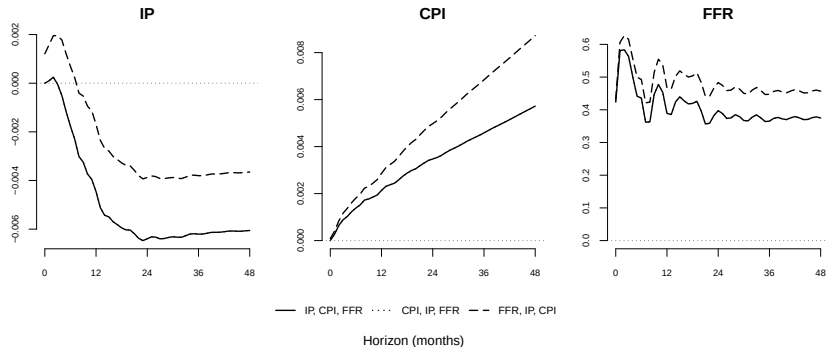
Structural Monetary-Policy IRFs

- With FFR ordered last, structural and reduced-form policy IRFs are proportional

$$\Theta_h e_3 = b_{33}^0 \Phi_h e_3$$

- Both curves have identical profiles; their magnitudes differ through shock normalization
 - IP declines, while the CPI level rises following a positive FFR shock
- ⇒ The price-puzzle pattern persists under recursive identification

Ordering Sensitivity



Ordering Sensitivity

- **FFR last:** Swapping IP and CPI leaves the monetary-policy IRFs unchanged
 - **FFR first:** IP and CPI can respond immediately to policy shocks, changing the estimated responses
 - Unit-variance structural shocks can also produce different FFR impact magnitudes across orderings
- ⇒ Ordering comparisons reveal sensitivity, but may represent different causal experiments

VAR/LP IRFs Comparison

Local Projections

- Local projections (Jordà, 2005) estimate IRFs directly through horizon-specific regressions

$$y_{t+h} = \alpha_h + \Psi_h y_t + \sum_{\ell=1}^p \Gamma_{\ell,h} y_{t-\ell} + v_{t+h}^{(h)}$$

- A separate regression is estimated for each horizon $h = 0, \dots, H$
 - The coefficients $\widehat{\Psi}_h$ give the reduced-form responses to unit innovations, holding the other contemporaneous innovations fixed
- $\Rightarrow$ IRFs are obtained directly, without iterating a fitted VAR

VARs versus Local Projections

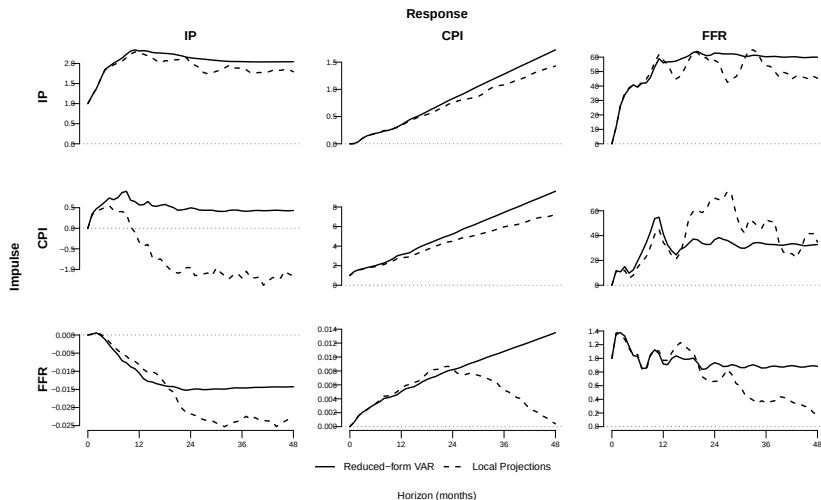
	VAR	Local Projections
Estimation	Estimates one dynamic system; IRFs are obtained by iterating its estimated coefficients	Estimates a separate regression at each horizon, without imposing a common dynamic structure
Advantages	Typically more precise and smoother IRFs, as information is shared across horizons	More flexible response shapes; often less bias when the VAR lag structure is misspecified
Disadvantages	Misspecified dynamics can introduce bias that propagates across horizons	Higher sampling variance, especially at longer horizons; potentially irregular IRFs

⇒ Both target the same population IRFs with sufficiently rich lag structures, but differ in their finite-sample bias-variance trade-off

Reduced-Form LP Implementation

- Estimate horizon-specific OLS regressions using `lm.fit()`
 - Same IP, CPI and FFR transformations, with 14 lagged controls and 48-month horizons
 - Contemporaneous regressors enter jointly, yielding nine unorthogonalized reduced-form IRFs
 - Accumulate responses once for IP and FFR, twice for CPI
- ⇒ VAR and LP estimates are compared for the same unit innovations

Reduced-Form IRFs: VAR versus LP



Reduced-Form IRFs: VAR versus LP

- All nine IRFs coincide on impact, with larger differences at longer horizons
 - LP estimates exhibit less smoothness and different persistence across the system
 - Following an FFR innovation:
 - IP falls under both estimators, with a larger long-horizon decline under LP
 - CPI initially rises under both, but the LP response subsequently declines toward zero while the VAR response continues rising
- ⇒ The price-puzzle pattern appears initially, but its persistence depends on the estimator

Structural IRFs Using Local Projections

Structural LP: Identification and Estimation

- LPs accommodate recursive, long-run, sign and external-instrument identification
- We adopt recursive identification with FFR last, matching the SVAR

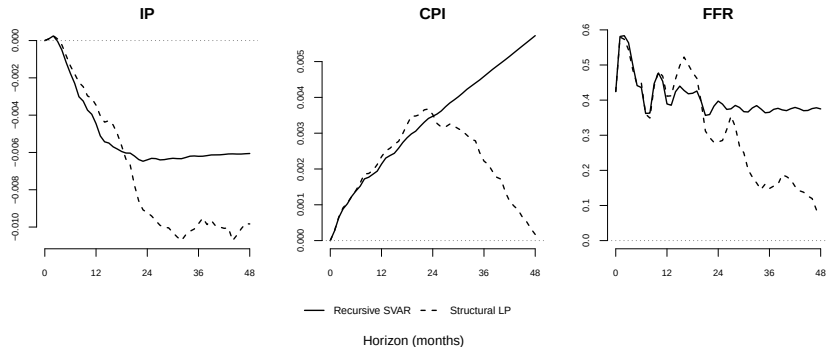
$$y_{j,t+h} = \alpha_{j,h} + \beta_{j,h} FFR_t + \gamma'_{j,h} (IP_t, CPI_t)' + \sum_{\ell=1}^{14} \delta'_{j,h,\ell} y_{t-\ell} + v_{j,t+h}^{(h)}$$

- Use `lp_lin()` with Cholesky identification and horizon-specific regressions

$$\widehat{\Theta}_{j,3,h}^{LP} = \widehat{b}_{33}^0 \widehat{\beta}_{j,h}$$

⇒ Same identifying assumptions and shock normalization, but direct LP estimation

Structural IRFs: SVAR versus LP



Structural LP: Interpretation

- Both estimators impose the same FFR-last recursive restrictions and agree on impact
 - LP responses exhibit different medium- and long-horizon dynamics relative to the SVAR
 - CPI initially rises under both methods, but the LP response subsequently weakens toward zero
- ⇒ Recursive identification does not remove the short-run price-puzzle pattern; its persistence depends on the estimator